

WICHITA FALLS SHEPPARD AFB, TX, USA

WMO: 723510

Lat: 33.979N

Lon: 98.493W

Elev: 310

StdP: 97.66

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13966

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.5	(c) -5.3	(d) -15.2	(e) 1.0	(f) -3.8	(g) -12.8	(h) 1.3	(i) -0.4	(j) 14.0	(k) 6.2	(l) 12.4	(m) 7.8	(n) 4.7	(o) 10	(p) 0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	39.9	22.6	38.4	22.7	36.9	22.8	25.4	33.4	24.9	32.9	24.3	32.3	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.8	28.5	22.7	18.1	27.8	22.2	17.5	27.4	79.6	33.6	77.0	32.9	74.8	32.4	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.2	(b) 11.0	(c) 9.8	(e) -11.2	(f) 41.8	(g) 2.1	(h) 1.8	(i) -12.7	(j) 43.1	(k) -14.0	(l) 44.2	(m) -15.1	(n) 45.2	(o) -16.7	(p) 46.5		
(4) 12.2	(5) 11.0	(6) 9.8	(7) -12.1	(8) 27.0	(9) 2.0	(10) 1.4	(11) -13.6	(12) 28.0	(13) -14.7	(14) 28.8	(15) -15.8	(16) 29.7	(17) -17.3	(18) 30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	6.2	8.2	12.9	17.5	22.5	27.1	29.5	29.4	25.0	18.5	12.1	6.7	(6)
(7)		DBStd	9.56	5.64	6.10	5.73	4.62	4.26	2.93	2.73	2.88	3.86	4.81	5.18	5.28	(7)
(8)		HDD10.0	448	144	96	36	4	0	0	0	0	0	3	36	128	(8)
(9)		HDD18.3	1541	377	285	183	69	13	0	0	0	3	57	193	361	(9)
(10)		CDD10.0	3377	26	46	126	228	388	513	606	600	449	268	100	27	(10)
(11)		CDD18.3	1430	1	2	14	43	143	263	348	342	203	63	7	1	(11)
(12)		CDH23.3	17480	13	41	180	477	1518	3147	4704	4488	2188	636	78	11	(12)
(13)		CDH26.7	9344	1	9	47	148	669	1670	2803	2659	1097	226	14	1	(13)
(14)	Wind	WSAvg	5.0	5.0	5.2	5.6	5.9	5.5	5.3	4.7	4.4	4.3	4.8	5.0	4.8	(14)
(15)	Precipitation	PrecAvg	727	28	37	52	71	96	84	45	65	87	73	41	38	(15)
(16)		PrecMax	1205	114	116	160	202	432	218	141	210	260	214	174	176	(16)
(17)		PrecMin	362	0	0	3	7	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	163	24	28	35	49	79	59	39	49	63	57	39	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	27.5	30.8	33.4	37.5	40.2	41.8	41.8	39.2	34.0	28.7	24.7	(19)
(20)			MCWB	13.8	15.0	16.5	18.2	22.5	22.6	22.3	22.2	21.9	20.3	18.2	14.3	(20)
(21)		2%	DB	21.4	24.1	27.6	30.3	34.8	37.7	39.8	39.9	36.5	31.5	25.6	21.0	(21)
(22)			MCWB	12.3	14.0	16.7	18.6	22.0	22.6	22.6	22.7	22.3	20.5	16.6	13.0	(22)
(23)		5%	DB	18.5	21.2	25.2	28.1	32.7	35.8	38.4	38.3	34.5	29.2	23.1	18.2	(23)
(24)			MCWB	10.7	13.0	15.7	18.0	21.6	22.8	22.8	22.9	22.1	19.4	15.5	11.7	(24)
(25)		10%	DB	15.8	18.3	22.5	26.1	30.4	34.1	36.7	36.7	32.6	27.0	20.8	15.5	(25)
(26)			MCWB	9.5	11.6	14.6	17.4	21.2	23.1	23.1	23.0	21.8	18.6	14.8	10.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	17.9	20.1	22.7	25.2	26.4	26.3	25.8	25.1	23.5	20.5	17.8	(27)
(28)			MCD B	20.9	22.7	26.7	27.8	32.0	33.2	34.2	34.6	32.6	29.5	25.0	21.4	(28)
(29)		2%	WB	14.3	16.3	18.7	21.2	24.0	25.3	25.3	25.1	24.1	22.4	18.5	15.1	(29)
(30)			MCD B	18.5	20.7	24.0	26.5	31.2	32.5	33.6	33.8	31.3	28.1	23.3	18.3	(30)
(31)		5%	WB	11.8	14.6	17.5	20.2	23.2	24.5	24.8	24.5	23.5	21.3	17.2	12.8	(31)
(32)			MCD B	17.0	19.6	22.5	25.3	29.9	31.9	33.1	33.2	30.5	26.3	21.4	17.0	(32)
(33)		10%	WB	9.8	12.3	16.0	19.0	22.3	23.9	24.1	24.0	22.8	20.3	15.8	10.3	(33)
(34)			MCD B	15.5	18.0	21.2	24.0	28.6	31.2	32.5	32.5	29.6	25.0	19.9	15.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.8	13.0	13.4	13.3	12.4	12.1	12.8	12.5	12.8	12.7	12.2	(35)	
(36)			MCDBR	17.4	17.5	16.9	16.0	14.9	14.4	14.6	14.7	14.6	15.2	15.6	16.5	(36)
(37)			MCWBR	9.8	9.5	8.5	7.8	5.9	4.3	3.7	4.0	4.4	6.8	8.4	9.7	(37)
(38)		5% WB	MCD BR	15.5	15.7	13.6	12.8	12.8	12.0	12.3	12.5	11.8	11.8	13.1	13.7	(38)
(39)			MCWBR	9.8	10.3	7.7	7.5	6.2	4.9	4.4	4.0	4.3	6.2	8.6	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.293	0.304	0.337	0.373	0.411	0.427	0.426	0.419	0.387	0.352	0.317	0.297	(40)	
(41)		taud	2.518	2.501	2.422	2.349	2.312	2.303	2.345	2.311	2.376	2.459	2.492	2.506	(41)	
(42)		Ebn at Noon	926	951	938	914	877	859	858	859	874	882	888	899	(42)	
(43)		Edh at Noon	82	93	110	124	130	131	125	127	114	97	84	79	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.00	3.59	4.77	5.80	6.22	6.89	6.90	6.32	5.25	4.08	3.16	2.57	(44)	
(45)		RadStd	0.31	0.52	0.30	0.47	0.65	0.59	0.43	0.36	0.44	0.58	0.40	0.29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	(47) Regional (41 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page